
Air Quality Prediction System

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Table of Contents

1. Problem Description 3

2. Architecture 4

3. Models Used..... 5

4. Evaluation metrics used6 - 7

5. Preliminary Results and Visualizations..... 8 - 9

6. References..... 10

1. Problem Description

Air pollution is a growing environmental concern that directly affects public health and quality of life. Monitoring air quality in real time and classifying it into meaningful health categories is essential for timely decision-making by governments and the general public. This project aims to automate the classification of Air Quality Index (AQI) readings into three categories — GOOD, Moderate, and WORST — using machine learning algorithms applied to sensor-collected pollutant data.

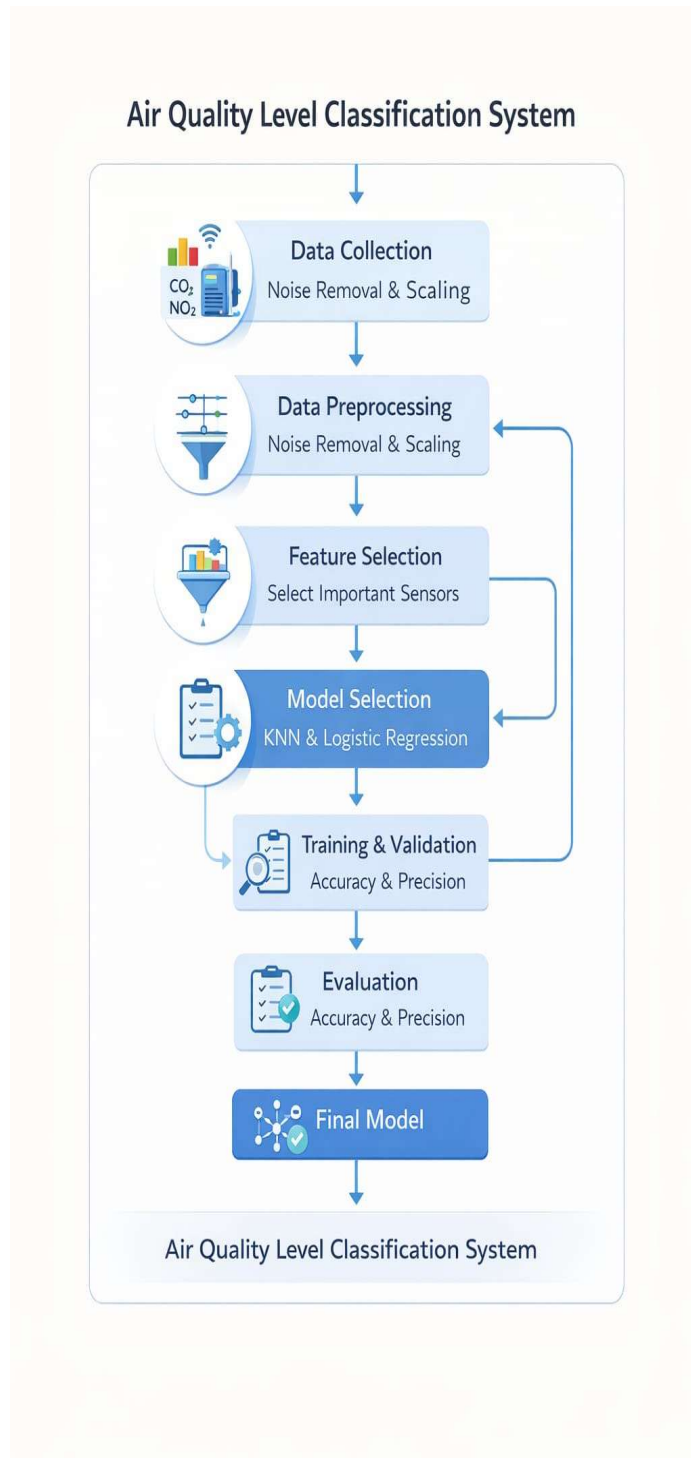
Dataset

The dataset consists of 5000 records with five attributes: Carbon Monoxide (CO), Nitrogen Dioxide (NO2), Temperature, Humidity, and Air Quality Index. CO and NO2 are the primary pollutants and act as the main drivers of the AQI value. The dataset was designed to reflect realistic sensor conditions and was intentionally injected with approximately 5% missing values per column, 2% outliers, and 1% duplicate records to simulate real-world data quality challenges.

Methodology

The project follows a complete machine learning pipeline. In the preprocessing phase, missing values were handled using median imputation, duplicate records were removed, and outliers were treated using the IQR clipping method. The AQI numerical values were then converted into three labeled categories. Features were standardized using StandardScaler, and Forward Sequential Feature Selection was applied to identify the most relevant features for distance-based models. Five classification models were trained and tuned using GridSearchCV with 2-fold cross-validation: K-Nearest Neighbors, Logistic Regression

2. Architecture



3.Models Used:

1. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a simple and intuitive supervised machine learning algorithm that classifies a data point based on the majority class of its K nearest neighbors in the feature space. It is a non-parametric and lazy learning algorithm, meaning it does not build an explicit model during training but instead stores the entire training dataset and computes distances at prediction time. In this project, KNN was applied to classify air quality into GOOD, Moderate, and WORST categories using Euclidean and Manhattan distance metrics. Hyperparameter tuning was performed using GridSearchCV to find the optimal value of K and the best distance weighting strategy.

2.Logistic Regression

Logistic Regression is a statistical and supervised machine learning algorithm primarily used for classification tasks. Despite its name, it is a classification algorithm that models the probability of a data point belonging to a particular class using the logistic (sigmoid) function. It is a linear model that works by finding the best decision boundary that separates the classes in the feature space. In this project, Logistic Regression was extended to handle the three-class classification problem — GOOD, Moderate, and WORST — using the multinomial approach with lbfgs and saga solvers. Regularization strength was tuned using GridSearchCV to prevent overfitting and improve generalization on the air quality dataset.

4. Evaluation metrics used

1. Accuracy

Accuracy is the most straightforward evaluation metric that measures the proportion of correctly classified samples out of the total number of samples. It gives a general overview of how well the model is performing across all classes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

In this project, accuracy was the primary metric used to compare all five models, with Random Forest achieving the highest accuracy of 87.84%.

2. Precision

Precision measures how many of the samples predicted as a particular class were actually correct. It focuses on the quality of positive predictions and is especially important when the cost of false positives is high.

$$Precision = \frac{TP}{TP + FP}$$

Weighted precision was calculated across all three classes — GOOD, Moderate, and WORST — to account for class imbalance in the dataset.

3. Recall (Sensitivity)

Recall measures how many of the actual samples belonging to a class were correctly identified by the model. It focuses on the model's ability to detect all relevant instances and is critical when missing a true positive has serious consequences.

$$Recall = \frac{TP}{TP + FN}$$

In the context of air quality classification, high recall for the WORST category is particularly important as missing a severely polluted condition could have serious public health implications.

4. F1-Score

F1-Score is the harmonic mean of Precision and Recall. It provides a single balanced metric that considers both false positives and false negatives, making it more informative than accuracy alone when dealing with multiple classes.

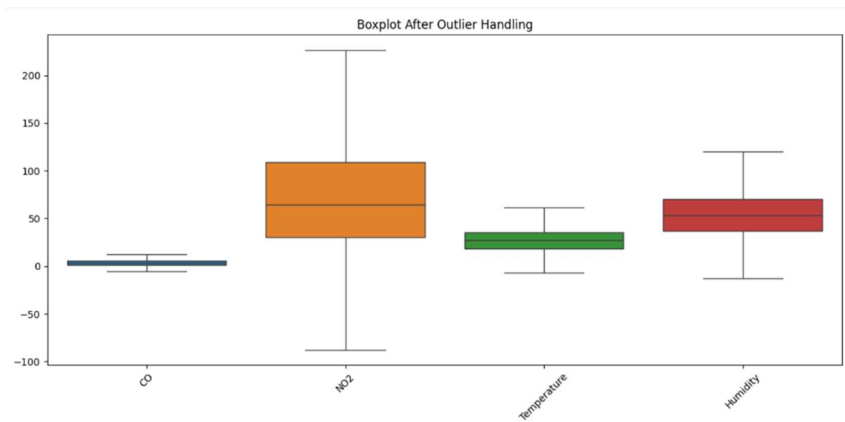
$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Both Weighted F1-Score (accounts for class size) and Macro F1-Score (treats all classes equally) were computed for all models in this project.

5. Classification Report

The Classification Report combines Precision, Recall, and F1-Score for each individual class into a single consolidated output. It provides both per-class metrics and overall weighted and macro averages, giving a complete picture of model performance across all three categories — GOOD, Moderate, and WORST.

5. Preliminary Results and Visualizations



KNN Accuracy: 0.8619 (86.19%)

	precision	recall	f1-score	support
GOOD	0.88	0.93	0.90	332
Moderate	0.81	0.78	0.79	319
WORST	0.90	0.87	0.88	319
accuracy			0.86	970
macro avg	0.86	0.86	0.86	970
weighted avg	0.86	0.86	0.86	970

Logistic Regression Accuracy: 0.8165 (81.65%)

	precision	recall	f1-score	support
GOOD	0.83	0.88	0.85	332
Moderate	0.73	0.76	0.74	319
WORST	0.91	0.81	0.86	319
accuracy			0.82	970
macro avg	0.82	0.82	0.82	970
weighted avg	0.82	0.82	0.82	970

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar
2. Scikit learn Documentation

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- Kaggle – <https://www.kaggle.com/>